

2. Rozkład wielomianu na czynniki 2

Wielomian

$w(x) = (x^2 - 1)(2x + 3) - (1 - x)(x + 1)^2 - (x^3 - x)(3x + 2)$ zapisano w postaci $w(x) = (x^2 - 1) \cdot p(x)$. Wskaz wielomian $p(x)$.

$$\begin{aligned}\omega(x) &= (x^2 - 1)(2x + 3) - (1 - x)(x + 1)^2 - (x^3 - x)(3x + 2) = \\&= (x^2 - 1)(2x + 3) + (x - 1)(x + 1)(x + 1) - x(x^2 - 1)(3x + 2) = \\&= \underline{(x^2 - 1)(2x + 3)} + \underline{(x^2 - 1)(x + 1)} - x \underline{(x^2 - 1)(3x + 2)} = \\&= (x^2 - 1)[(2x + 3) + (x + 1) - x(3x + 2)] = \\&= (x^2 - 1) \cdot \underbrace{(-3x^2 + x + 4)}_{p(x)}\end{aligned}$$

Odp: $p(x) = -3x^2 + x + 4$.